SEMINAR REPORT · 7 MINUTES

Reverse typography: three seminars

OCR in the age of VLMs, the printer behind a book, segmentation we can explain — read through a programme of our collaborators.



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WHAT I FOLLOWED

Three seminars, one thread

SEMINAR · 10 OCT 2025

What is OCR really worth today?

De la vision par ordinateur aux LLMs : que vaut vraiment l'OCR aujourd'hui ?

ILB Data Lab × Centre Borelli × IRISA, one hour online.
S. Mowlavi, M. Gardella (Centre Borelli)

GTTI SEMINAR · 26 MAR 2026

Analyzing historical documents beyond text transcription

D. Belzarena.
Comedias sueltas: Spanish Golden-Age plays printed as cheap pamphlets. Who printed them?

DEFENSE · 6 OCT 2026

Mathematical formalization of image segmentation

R. W. Leon (PhD, supervised by J.-M. Morel and Z. Liu).
Train on data we fully specify, so we can explain what the model learns.

Collaborators: Seginus Mowlavi and Marina Gardella (Centre Borelli), and Diego Belzarena. For each talk: what it does, one strength, one open question.

Reverse typography: run the press backwards

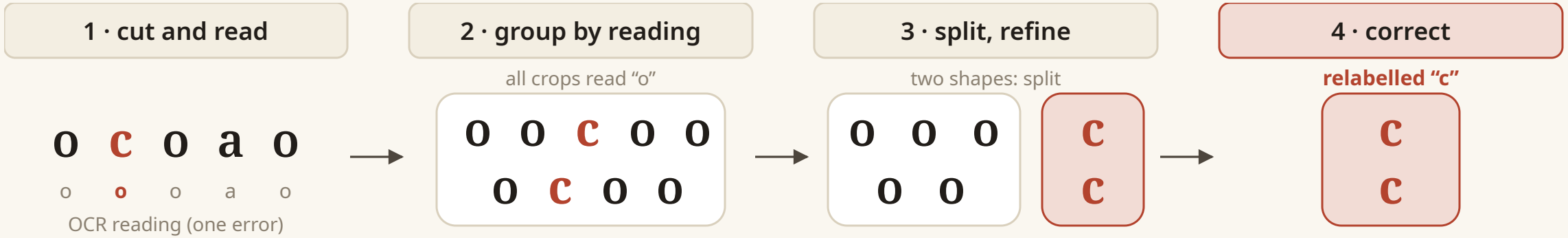


Definition. Extract the shapes of the printed characters from scanned documents, recognise them, and restore them in a printable form. J.-M. Morel, "Reverse typography and the theory of shape: can old books be brought back to life?", FAU MoD lecture, 19 Jun 2026.

- **Why:** a history of typography from the books themselves; old books re-edited with their original appearance.
- **How:** the mathematical theory of planar shapes and its PDEs — where Leon's Mumford–Shah formulation sits.
- Transcription is one stage, not the goal.

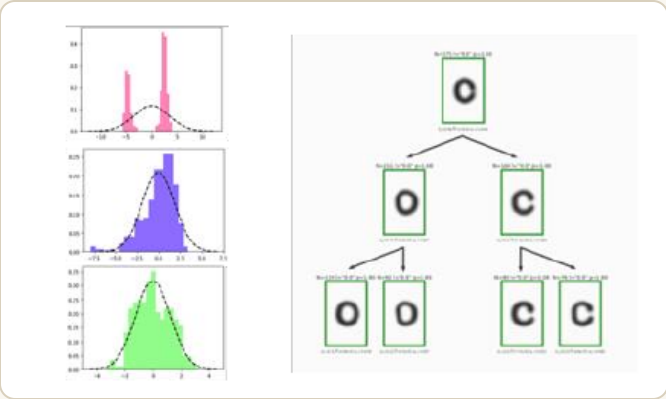
Learn the fonts of a corpus, without labels

Why it can work. In a printed book, every “a” comes from a few metal sorts cast in one mould: the same shape, repeated hundreds of times. OCR trained elsewhere fails on old prints, and annotating each book costs too much — so exploit the repetition instead.



The result is the book’s own font: one prototype per letter shape — the font stage of reverse typography.

What the clusters give: a book's own font



Two humps in a group's statistics (pink): two shapes, split. One hump: keep. "o" splits into "o" and "c", then into variants. His slides; my reading of them.



Two editions of one *comedia*, their letters side by side: the two fonts can be compared. His slides.

- **OCR correction** with no annotation: each group must agree.
- **A model of the font**: one prototype per letter shape.
- **Philology**: compare editions, letter by letter (the bridge to Belzarena).
- **Forensics**: a character that fits no group of its page is suspect.

STRENGTH The redundancy of print is the only supervision, and a group can be looked at: an error is visible.

OPEN QUESTION When is a split a genuinely new type (another font, a broken sort) and not noise? How is a letter seen five times corrected?

Who printed these plays? The page must say



Title pages of five *sueltas*; colours group the same piece of type across books. D. Belzarena's slides.

40,952 of the **78,524** volumes printed in 17th-c. Spain name no printer, often to evade legal controls. *Sueltas*: plays in quarto, popular, short, on cheap paper.

≈ 5,000 detailed pairwise comparisons an expert would need for about 100 books.

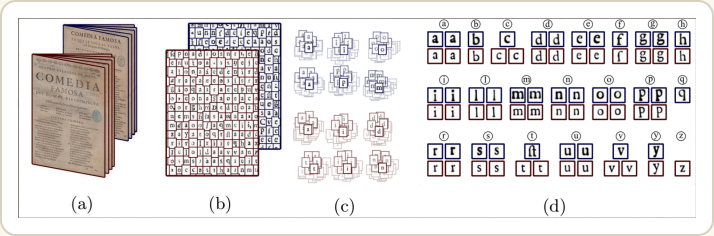
- **Today, by hand**: compare a book's typefaces, ornaments and page layout with books of certain origin; build a network of typographic relations.
- **The trap**: same typeface ≠ same printer — type was traded and shared between shops.
- **Other clues**: watermarks, spacing and spelling habits (Shakespeare's First Folio), damaged sorts.

Reverse typography's question. From the printed page back to the metal type that printed it — and from the type to the print shop.

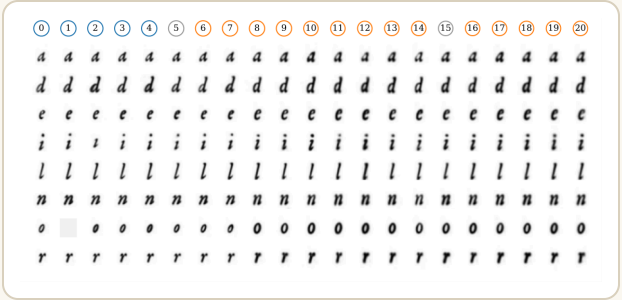
Recover each book's type, then count what differs

Reverse typography in practice. Each book's letters are traced back to the type that printed them: one prototype per piece of type. Books are then compared type by type, not text by text.

1. **Find the characters** (a CharNet detector) and sort roman from italic by the slant of each word.
2. **One prototype per type, per book**: cluster the book's characters and average each cluster — the clustering of Mowlavi's talk.
3. **Compare two books letter by letter**: align the two prototypes of each letter, take their distance (the closest pair, since a book may own several variants of a letter).
4. **Count, a contrario, the letters that surely differ**: if the books shared their type, chance alone would make few distances large (a binomial law). The smallest count compatible with it, at 0.01 false alarms over all pairs, is $\hat{n}_1$; the share of differing letters links the books.

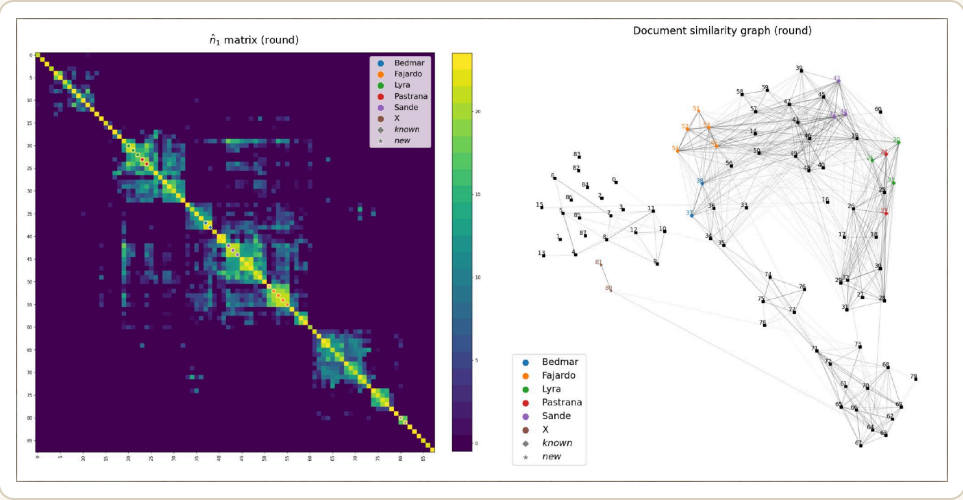


(a) books, (b) characters, (c) clusters per book, (d) prototypes matched by letter.



Italic prototypes, one column per book: books 0–7 share one italic (thin "o"), 8–20 another. Belzarena, Mowlavi et al., 2026, fig. 6.

New attributions, confirmed by experts



88 *sueltas* (BNE): books sharing type are linked; colours mark the 14 with a known printer (Bedmar, Fajardo, Lyra, Pastrana, Sande). His slides.

0.99 AUC for same- vs different-printer on the 91 pairs of known books (roman 0.989, italic 0.997).

book 51 joins Fajardo's family; experts then found shared layout, letterhead type and type damage.

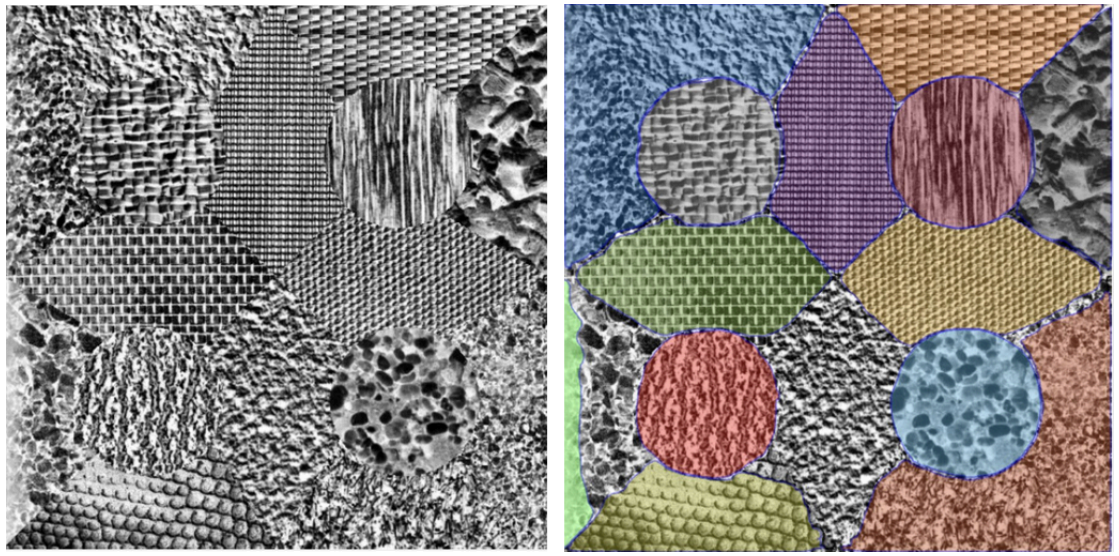
80, 81 a new printing family, perhaps a century apart from the others.

6, 7 moved from Lyra to Rodríguez de Ábrego (their italic, a broken title type); two unattributed books placed.

STRENGTH No training data, a statistical guarantee on every link, and output that experts check by the classical method: about 15 minutes per book instead of thousands of comparisons.

OPEN QUESTION Shared type is not a shared shop: what separates them beyond type (layout, ornaments, damage)? And with no ground truth beyond 14 books, how far does validation scale to the 623 sueltas already collected?

A great model we cannot explain

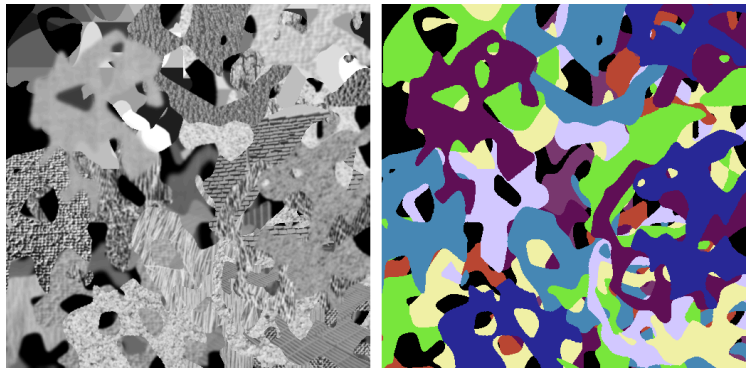


14 Brodatz textures, exact boundaries; SAM's masks, grey = missed: 4 of 14 textures get none. R. W. Leon's slides.

- **SAM**: 11M images, 1.1B masks — but natural photos come with no specification, so its failures can be neither explained nor fixed through the data.
- **Before deep learning**, a segmentation minimised an explicit energy (Mumford–Shah), whose existence and regularity can be proved: $E(\mathrm{Seg}) = \sum_R \mathrm{var}(g, R) + \mu\,\mathrm{length}(\partial R)$.
- **Question**: do segmentation networks learn *objects*, or low-level features — colour, texture, geometry?

Specify the data, explain the model

For reverse typography, the shape stage: segmentation defined by an explicit energy, learned where the truth is exact — a glyph generator would run the same experiment on print.



Each image is a seed plus recorded variables: scene, shape, colour, texture, acquisition. R. W. Leon's slides.

step	what	status
generator	dead leaves (from VibrantLeaves), real Brodatz / ALOT textures, difficulty presets	done
Abstract SAM	encoder from scratch, prompt-free decoder, boundary + Mumford–Shah-like loss	running
benchmark	levels L1–L5, up to natural images; SAM in automatic mode	planned
explain	one variable at a time, ablations, the same tests on SAM	planned

STRENGTH An observational study turned into a controlled experiment: every error traces to a variable, and he names the bias of his own evaluation (SAM given oracle prompts).

OPEN QUESTION External validity (level L5): will a model trained on leaves segment real images? No results yet to judge.

SYNTHESIS

What recurs, and what stays open

STRENGTHS

- **Reverse typography**: each talk inverts one stage of printing — the font (Mowlavi), the type and the shop (Belzarena), the shape (Leon). Results can be checked.
- **Frugal supervision**: a book's own redundancy, or fully specified synthetic data, replaces annotation.
- **Explanations, not only scores**: clusters one can look at, an a contrario guarantee on every link, an explicit energy.

OPEN QUESTIONS

- **Validation**: only 14 books with a known printer; no results yet for the segmentation model.
- **Transfer**: from 88 to 623 *sueltas*; from leaves to real images, and to glyphs.
- **What a difference means**: another printer, another piece of type, or noise?
- **One inversion**: font, type and scan recovered jointly, to check what a fluent VLM reads.

Reverse typography: when labels are scarce, invert how the page was printed.

THANK YOU

Questions?

Reverse typography: from the printed page back to the type, the press and the print shop.

BACKUP

Appendix

For questions. Not counted in the seven minutes.

BACKUP

Echoes in my own work

seminar	idea	in my project (historical Chinese prints, TKH)
Mowlavi	cluster characters, correct by agreement	a k-NN graph over every glyph; labels corrected by diffusion on it
Gardella	a contrario NFA	graph edges kept by an a contrario test on glyph distances
Belzarena	Learnable Typewriter	ported, trained on TKH: test CER 1.44 % vs 8.87 % for CHAT; characters seen < 5 times are never read
Leon	specified synthetic data	glyph boxes from CRAFT refined by the SAM 3 tracker; a font-rendered generator would make glyph segmentation a controlled experiment

BACKUP

References cited in the seminars, and the frame

[1] T. Berg-Kirkpatrick et al. *Unsupervised Transcription of Historical Documents.* 2013.

[2] T. Berg-Kirkpatrick et al. *Improved Typesetting Models for Historical OCR.* 2014.

[3] M. Ryskina et al. *Automatic Compositor Attribution in the First Folio of Shakespeare.* 2017.

[4] N. Vogler et al. *Contrastive Attention Networks for Attribution of Early Modern Print.* 2023.

[5] I. Siglidis et al. *The Learnable Typewriter: A Generative Approach to Text Analysis.* 2024.

[6] D. Belzarena, S. Mowlavi, P. Casariego Castiñeira, A. Ulla Lorenzo, G. Randall, J.-M. Morel. *Theatre Chapbooks At Scale: A Statistical Comparative Analysis of Typography.* arXiv:2607.27266, 2026.

[7] J.-M. Morel. *Reverse typography and the theory of shape: can old books be brought back to life?* FAU MoD lecture, 19 Jun 2026.

Also named, without a reference list: SAM, Mumford–Shah, VibrantLeaves, Brodatz and ALOT textures (Leon).